

BADIS MOALLA

Software Test & Validation Engineer | Embedded Systems | Diagnostic Protocols | Process Improvement

Tunis, Tunisia | Requires work permit sponsorship

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PROFESSIONAL SUMMARY

Software Test and Validation Engineer with 3+ years at KPIT Engineering within the BMW Group supply chain. Specialist in embedded ECU validation, diagnostic protocol testing (UDS, DoIP, DLT analysis), and security domain verification on AUTOSAR Adaptive platforms. Experienced in Python-based test automation and data-driven process improvement. Operated within an ISO 26262 and ASPICE-aligned V-model environment. Trilingual: Arabic, English and French (C1). Requires work permit sponsorship for Netherlands, Luxembourg, or Poland.

TECHNICAL SKILLS

Testing and Validation: Manual and automated software testing, embedded systems validation, test plan design, test case execution, defect lifecycle management, regression testing, traceability matrices

Embedded and ECU: AUTOSAR Adaptive, ECU certificate installation, token verification, ECU access modes, secure boot, DTC validation, diagnostics job testing

Diagnostic Protocols: UDS (ISO 14229), DoIP (ISO 13400), TCP/IP, SOME/IP (familiar), DLT log analysis, Wireshark, BMW Zedis diagnostic database

Standards and Process: ISO 26262 functional safety (aware), ASPICE V-model (SWE.6), root-cause analysis, problem management, KPI tracking, Agile, Scrum

Test Management Tools: Test Guide (TG), Jira (dashboards, workflow design, API automation), Git

Automation and Data: Python (Pandas, NumPy, Matplotlib), Power BI (DAX, Power Query), SQL, Excel (Power Query, Pivot Tables)

Languages: Arabic (Native), English (C1 - Fluent), French (C1 - Fluent), German (Basic - A2)

PROFESSIONAL EXPERIENCE

Problem Manager - Process Improvement and Quality

KPIT Engineering (automotive supplier to BMW Group), Sfax, Tunisia | May 2025 - Present

- Reduced repeat incident rate by an estimated 30% over two quarters by mediating root-cause sessions between development and testing teams and implementing structured process fixes.
- Improved real-time management visibility by designing Jira dashboards from scratch tracking team deviations, SLA compliance, and backlog health across two project teams.
- Cut manual reporting effort by 40% by writing Python scripts to automatically extract, clean, and structure Jira deviation data, replacing 3-4 hours of weekly manual work.
- Prevented client escalations by managing end-to-end reclamation handling: investigated failure origin across 100,000+ records, implemented corrective process changes, and confirmed closure with stakeholder sign-off.
- Reduced incident resolution time by 25% by building a Power BI dashboard correlating defect patterns with release timelines, enabling data-driven prioritisation of high-recurrence fixes.

Software Test and Validation Engineer - ECU Security and Diagnostics Technical Reference

KPIT Engineering (automotive supplier to BMW Group), Sfax, Tunisia | Feb 2023 - May 2025

- Ensured BMW ECU security compliance by serving as technical reference for the security and diagnostics test domain, designing full validation coverage for certificate installation, token verification, and ECU access mode sequences.
- Detected 100+ diagnostic defects across multiple ECU variants by validating DTC and diagnostics job responses using UDS (ISO 14229) and DoIP (ISO 13400) protocols, dissecting service frames against BMW Zedis diagnostic database specifications.
- Reduced average defect reproduction time from days to hours by analysing DLT log frames and Wireshark packet captures on AUTOSAR Adaptive ECU platforms, providing developers with precise failure context.
- Maintained zero traceability gaps across all release cycles by managing test data configurations and traceability matrices in Test Guide (TG), ensuring full auditability for BMW stakeholder reviews.

- Increased test coverage visibility by building Power BI dashboards tracking defect rate, test coverage, and release quality KPIs, referenced in weekly project reviews with BMW client representatives.
- Contributed to ASPICE process compliance by executing software qualification testing (SWE.6) within a V-model development lifecycle, producing structured work products reviewed by BMW stakeholders.

Data Analyst and BI Developer - Intern

AT Worthy (Remote, Virginia, USA) | Mar 2022 - Oct 2022

- Improved executive decision-making by designing Power BI and Tableau dashboards integrating data from multiple business systems into a unified reporting view.
- Reduced manual data preparation effort by building ETL pipelines using SSIS for automated extraction, transformation, and loading across heterogeneous data sources.

BI Developer - Intern

Clinisys, Sfax, Tunisia | Jun 2020 - Sep 2020

- Enabled real-time operational monitoring by building Power BI and Python dashboards that replaced a weekly manual reporting process for the operations team.
- Reduced ad-hoc data request turnaround from days to hours by designing reusable SQL scripts for automated reporting and data extraction.

KEY PROJECTS

ECU Diagnostics Deviation Tracker | Python, Jira API, Power BI, DLT, Wireshark

- Reduced incident resolution time by 25% by analysing 6 months of ECU defect data (50,000+ records) to identify top diagnostic failure root causes and building an interactive Power BI dashboard used in BMW project reviews.
- Eliminated ambiguous defect classification by correlating Jira deviation records with DLT log analysis, providing developers with precise failure context that cut average debug time by an estimated 2 hours per issue.

Automated Deviation Reporting Pipeline | Python, Jira API, Excel

- Saved 3-4 hours of manual work per week by building a Python pipeline that automatically extracted Jira deviation data, applied cleaning rules, and generated structured client-ready reports.
- Improved reporting consistency by standardising the deviation extraction logic into a reusable script adopted as the team standard for client reporting.

EDUCATION

Diploma in Industrial Computer Engineering (Bac+5 - Master's Level Engineering Degree)

ENET'Com - National School of Electronics and Telecommunications, Sfax, Tunisia
Sep 2019 - Dec 2022

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Python Developer Certificate
- Scrum Fundamentals Certificate
- Linux Fundamentals Certificate
- Microsoft Power BI Data Analyst (PL-300) - In progress (2026)